



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE (Advanced)

SCORE - II

01-05-2021

JEE(Main + Advance) : Enthusiast Course (PHASE : SCORE-II)

Test Type : SCORE - II

Test Pattern : JEE (Advanced)

TEST DATE : 01-05-2021

ANSWER KEY

PAPER-2

PHYSICS

SEC-1	Q.	1	2	3	4	5	6	7
	A.	C	B	D	D	D	A	D
SEC-2	Q.	1	2	3	4	5	6	7
	A.	A,B	B,D	A,C,D	A,C	A,C	A,C	B,D
SEC-3	Q.	1	2	3	4			
	A.	C	C	B	C			

CHEMISTRY

SEC-1	Q.	1	2	3	4	5	6	7
	A.	C	A	A	D	D	C	C
SEC-2	Q.	1	2	3	4	5	6	7
	A.	A,C,D	A,B,C	A,B	B,C	B,C,D	A	B,D
SEC-3	Q.	1	2	3	4			
	A.	A	C	A	B			

MATHEMATICS

SEC-1	Q.	1	2	3	4	5	6	7
	A.	B	A	C	C	C	A	B
SEC-2	Q.	1	2	3	4	5	6	7
	A.	A,B	A,B	A,B,C,D	A,B	A,B,C	B,C	A,B
SEC-3	Q.	1	2	3	4			
	A.	C	A	B	B			

HINT – SHEET

PHYSICS

SEC-1

1. **Ans (C)**

$$\tau_{\text{net}} = 0$$

About pivot = 0

2. **Ans (B)**

$$v_p = -v \times \text{slope}$$

$$x = \pi/8 \text{ at } t = \pi/6$$

Particle is at mean position

3. **Ans (D)**

$$2\pi \rightarrow \lambda$$

$$\frac{\pi}{4} \rightarrow \frac{\lambda}{8}$$

$$\lambda = 40$$

$$L = 6 \left(\frac{\lambda}{2} \right) = 120$$

5. **Ans (D)**

$$LC \frac{1\text{mm}}{200}, \frac{0.5\text{mm}}{400}, 0.1 \text{ mm}, 0.2 \text{ mm}$$

6. Ans (A)

$$\beta = \frac{\lambda D}{d}$$

$$\Rightarrow \lambda = \frac{\beta d}{D} = \frac{2.50 \times 10^{-3} \times 3.3 \times 10^{-4}}{1.50} = 550$$

nm

$$\frac{\Delta \lambda}{\lambda} = \frac{\Delta \beta}{\beta} + \frac{\Delta d}{d} + \frac{\Delta D}{D} =$$

$$\frac{0.01}{0.33} + \frac{0.01}{2.50} + \frac{10^{-3}}{1.500}$$

$$\Delta \lambda = \frac{5.50}{0.33} + \frac{5.5}{2.5} + \frac{0.5.5}{1.50}$$

$$= \frac{500 + 66 + 11}{30} = \frac{577}{30}$$

7. Ans (D)

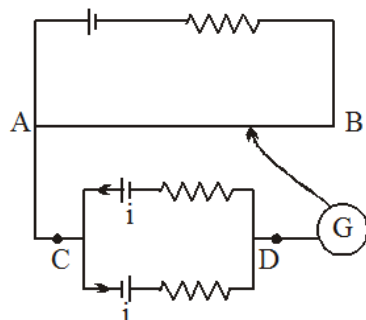
$$(S_1 O)_{\text{optical}} = \mu_1 (S_1 O)$$

$$(S_2 O)_{\text{optical}} = \mu_1 (S_2 O - t) + \mu_2 t$$

PHYSICS

SEC-2

1. Ans (A,B)



For null point current flow in the loop CD only.

$$i = \frac{3V}{2\Omega + 1\Omega} = 1 \text{ A}$$

$$V_{CD} = 1V - 1(1) = 0$$

$\therefore$ option (A) is correct is $V_A > V_B$

When Jockey touches to B current flow from A to B to increase the P.D. across the secondary circuit.

$\therefore$ option (B) is correct

2. Ans (B,D)

Applying Energy conservation, initially, kinetic energy = 0

gravitational P.E. = 0 (say) & Magnetic P.E. = 0

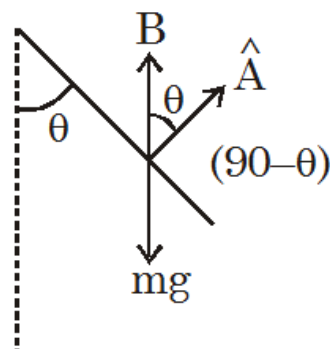
where, μ = magnetic moment of the loop

$$= i \cdot \left(\frac{\sqrt{3}a^2}{4} \right)$$

Finally when the loop becomes horizontal,

Kinetic energy = 0

gravitational P.E. = $mg \left(\frac{a}{\sqrt{3}} \right)$ (because mg acts on the centre of mass)



magnetic P.E. = $-\mu B$

$$\Rightarrow 0 + 0 + 0 = 0 + \frac{mga}{\sqrt{3}} - \mu B$$

$$\Rightarrow B = \frac{mga}{\sqrt{3}\mu} = \frac{4mg}{3ia}$$

for equilibrium position, torque due to magnetic force should be balanced by the torque due to gravitational force (mg).

$$\tau_B = \mu B \sin (90 - \theta);$$

$$= \frac{mg \sin \theta a}{\sqrt{3}}$$

3. **Ans (A,C,D)**

(A) E at C due to outside charges =

$$\frac{K2Q}{(2R)^2} = \frac{KQ}{2R^2}$$

(C) E at C due to induced charges =

$$\frac{Kq^2}{2R} + \frac{Kq'}{R} = 0$$

$$q' = -q$$

$$(D) \Delta Q = 4q - (-q) = 5q$$

4. **Ans (A,C)**

$$m\vec{v}_B + 2m\vec{v}_{AC} = 0$$

$$\vec{v}_{AC} = \frac{-\vec{v}_B}{2}$$

5. **Ans (A,C)**

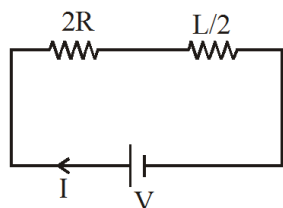
$$A_1V_1 = A_2V_2$$

$$W = (0.1)(10)(0.05) + (0.02)(10) \left(\frac{0.05}{2} \right) + \frac{1}{2} \left(\frac{20}{1000} \right) (2)^2$$

$$W = 0.095$$

6. **Ans (A,C)**

Given circuit can be reduced to given below.



$$\rightarrow I = I_0 \left(1 - e^{\frac{-R_{eq}t}{L_{eq}}} \right)$$

$$\text{or } I = \left(\frac{V}{2R} \right) \left(1 - e^{\frac{-2Rt}{L/2}} \right) \quad \text{or}$$

$$I = \frac{V}{2R} \left(1 - e^{\frac{-4Rt}{L}} \right)$$

$$\rightarrow I_1 = I_2 = \frac{I}{2} = \frac{V}{4R} \left(1 - e^{\frac{-4Rt}{L}} \right)$$

Since a and b are directly connected to battery, so the potential difference between a and b is constant.

$$\text{Thus } \frac{dV_{ab}}{dt} = 0 \quad \text{at } t = 0$$

7. **Ans (B,D)**

$$204 = -13.6 z^2 \left(\frac{1}{1} - \frac{1}{4n^2} \right)$$

$$40.8 = -13.6 z^2 \left(\frac{1}{n^2} - \frac{1}{4n^2} \right)$$

$$5 = \frac{1 - \frac{1}{4n^2}}{\frac{1}{n^2} - \frac{1}{4n^2}} \Rightarrow n = 2 \Rightarrow z = 4$$

$$E_1 = -13.6 \frac{(16)}{(1)^2} = -217.6 \text{ eV}$$

PHYSICS

SEC-3

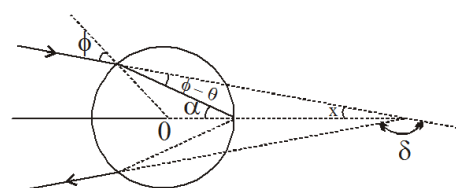
1. **Ans (C)**

$$W = P_0 V_0 - \frac{\pi P_0 V_0}{4}$$

2. **Ans (C)**

$$n = \frac{W}{Q_{\text{input}}} = \frac{0.22P_0 V_0}{P_0 V_0 + 0.22P_0 V_0 + 4.5P_0 V_0}$$

4. **Ans (C)**



(4) As $\alpha = (\phi - \alpha) + x$, or $x = 2\alpha - \phi$, we have $\delta = \pi - 2x = \pi - 4\alpha + 2\phi$.

(5) For minimum deflection, we require $\frac{d\delta}{d\phi}$

$$= -4 \frac{d\alpha}{d\phi} + 2 = 0, \text{ or } \frac{d\alpha}{d\phi} = \frac{1}{2}$$

$$\text{As } \alpha = \sin^{-1} \left(\frac{1}{n} \sin \phi \right)$$

we have $\frac{d\alpha}{d\phi} = \frac{1}{n} \frac{\cos \phi}{\cos \alpha}$ and the above gives

$$1 - \frac{1}{n^2} \sin^2 \phi = \frac{4}{n^2} \cos^2 \phi,$$

$$\text{or } 1 = \frac{1}{n^2} + \frac{3}{n^2} \cos^2 \phi,$$

$$\text{giving } \cos^2 \phi = \frac{n^2 - 1}{3}$$

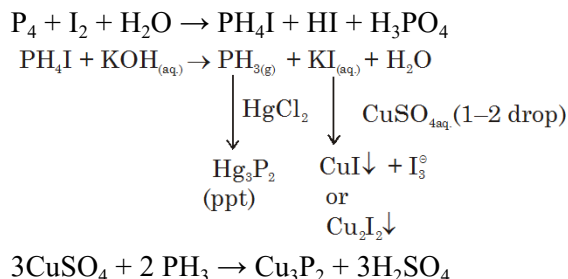
CHEMISTRY

SEC-1

3. Ans (A)

Option (A) is wrong because 'b' in trans position w.r.t. 'a'.

4. Ans (D)

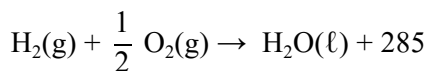
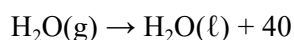
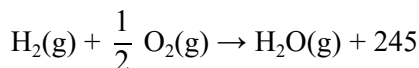


5. Ans (D)

$$E_{\text{cell}} = 1.23 - \frac{0.06}{2} \times \log \frac{1}{1 \times 0.25^{0.5}}$$

$$E_{\text{cell}} = 1.23 - 0.03 \times 0.3 = 1.221V$$

$$W = 2 \times \frac{96500}{1000} \times 1.221 = 235.653 \text{ kJ}$$



$$|\Delta_c H| = 285 \text{ kJ}$$

$$\eta = \frac{235.653}{285} \times 100 = 82.68\%$$

6. Ans (C)

(A) Brownian movement is responsible for the stability of colloidal solution

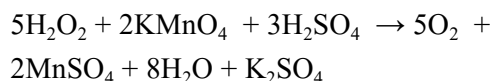
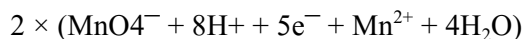
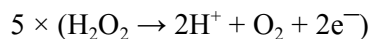
(B) Due to small size of particle in true solution, light does not scattered by the particle, so

no tyndall effect observed in true solution

(C) Haemoglobin positive charge sol, Gold sol negative charge sol

(D) Casein act as stabilising agent i.e. stabliser

7. Ans (C)



(A) No. of gm equivalent of H_2O_2 = no. of gm equivalent of $KMnO_4$

$$(B) \frac{\text{Moles of } MnSO_4}{\text{moles of } O_2} = \frac{2}{5}$$

$$(C) E = \frac{98}{10} \times 3 = 29.4$$

$$(D) x_{H_2O_2} = x_{KMnO_4}$$

$$\frac{2 \times 100 \times 0.2}{1000} = \frac{5 \times 0.5V}{1000}$$

$$\Rightarrow V = 16 \text{ ml.}$$

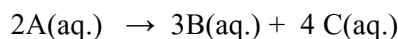
CHEMISTRY

SEC-2

3. Ans (A,B)

Size of cation decreases, size of anion increases, covalent character increases.

4. Ans (B,C)



$$a \quad 0 \quad 0 \quad t = 0$$

$$a - x \quad \frac{3x}{2} \quad 2x \quad t = 22 \text{ min}$$

$$0 \quad \frac{3a}{2} \quad 2a \quad t = \infty$$

$$a \times 20 = 120 - \theta \quad \dots\dots (i)$$

$$1.5 \times 40 + 2a \times (-80) = -120 - \theta \quad \dots\dots (ii)$$

$$(i) - (ii)$$

$$20a - 60a + 160a = 240$$

$$120a = 240 \Rightarrow a = 2$$

$$\theta = 80$$

$$(a - x) \times 20 + 1.5x \times 40 + 2x(-80) = -40 - \theta$$

$$40 - 20x + 60x - 160x = -40 - D$$

$$-120x = -80 - 80 = -160$$

$$x = \frac{160}{120} = \frac{4}{3}$$

$$K_A = \frac{1}{2} \ln \frac{2}{2 - \frac{4}{3}} = \frac{\ln 3}{22} = \frac{1.1}{22}$$

$$K_A = 0.05 \text{ min}^{-1}; K_C = 0.1 \text{ min}^{-1}$$

Time in which 50% of A.

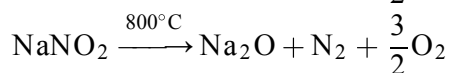
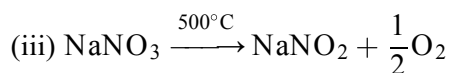
$$\text{reacted} = \frac{\ln 2}{K_A} = \frac{0.7}{0.05} = 14 \text{ min}$$

Time in which 75% of A reacted is $= 2 \times 14 = 28 \text{ min}$.

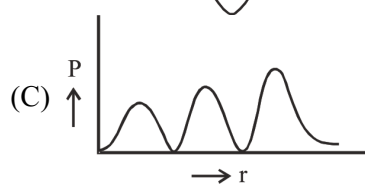
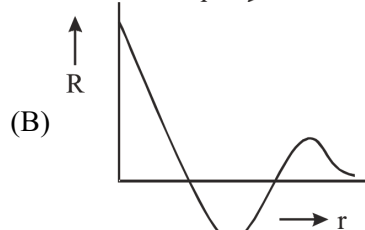
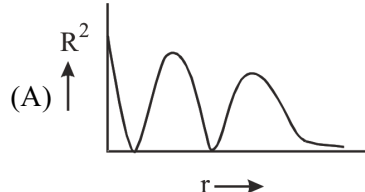
6. Ans (A)



(ii) Li^+ does not form alum because of its small size.



7. Ans (B,D)



$$(D) \sigma^2 - 4r + 3 = 0$$

$$\sigma^2 - 3r - r + 3 = 0$$

$$\sigma(\sigma - 3) - 1(\sigma - 3) = 0$$

$$(\sigma - 3)(r - 1) = 0$$

$$\sigma = 1, 3 \quad r = \frac{2r}{a_0}$$

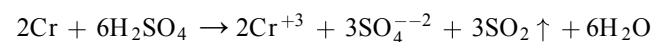
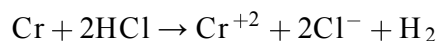
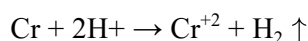
$$r_1 = \frac{a_0}{2} \quad r_2 = \frac{3a_0}{2}$$

$$d = r_2 - r_1 = \frac{3a_0}{2} - \frac{a_0}{2} = a_0 = 0.529 \text{ \AA}$$

CHEMISTRY

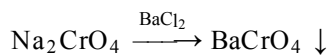
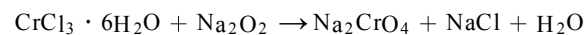
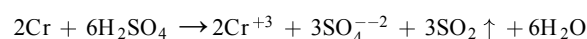
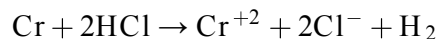
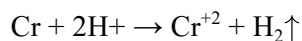
SEC-3

1. Ans (A)

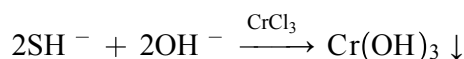
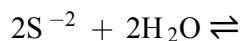
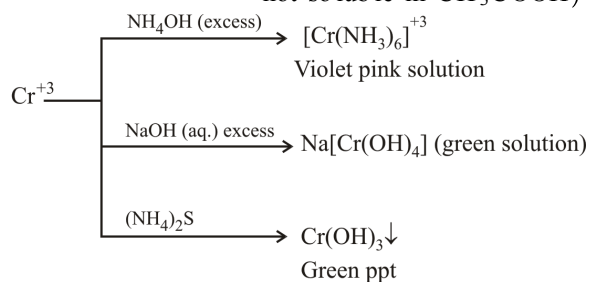


Colour of $\text{CrCl}_2(\text{aq.})$ and $\text{CrCl}_3(\text{aq.})$ are blue and green respectively

2. Ans (C)



Yellow ppt
(Soluble in dil. HCl but
not soluble in CH_3COOH)



MATHEMATICS

SEC-1

1. Ans (B)

$$P\left(\frac{E_2}{E_1}\right) = \frac{P(E_2 \cap E_1)}{P(E_1)}$$

$$\Rightarrow P(E_2 \cap E_1) = \frac{1}{8}$$

$$\Rightarrow P(E_2) = \frac{\frac{1}{8}}{P(E_1/E_2)} = \frac{1}{2}$$

$$\Rightarrow P(E_1)P(E_2) = \frac{1}{4} \times \frac{1}{2}$$

$$\Rightarrow P(E_1 \cap E_2) = P(E_1) \times P(E_2)$$

$$\text{Now, } P(E_1 \cup E_2) = \frac{5}{8}$$

so, E_1 and E_2 are independent but not exhaustive set of events.

2. Ans (A)

Case I : When 9, 9, 9 are used

$$\begin{array}{|c|c|c|c|c|} \hline 9 & & & & \\ \hline \end{array}$$

$$\text{Total number} = {}^9\text{C}_2 \times \frac{4!}{2!} = 432$$

Case II : When 3 identical digit is different from 9

$$\begin{array}{|c|c|c|c|c|} \hline 9 & & & & \\ \hline \end{array}$$

$$\text{Total number} = {}^9\text{C}_2 \times \frac{4!}{3!} \times 2 = 288$$

$$\text{So, } N = 720$$

4. Ans (C)

$$\text{Required number} = \sum_{c=0}^9 (10-c)(9-c) = 330$$

$$\text{As } b > c \text{ and } d \geq c$$

5. Ans (C)

$$B^4 = B \Rightarrow |B| = 1$$

$$|A| \neq |B| \Rightarrow |A| \neq 1$$

$$AA^T = I \Rightarrow |A| = -1$$

using properties of adjoint

$$\sum_{r=1}^{20} \det(\text{adj}(A^r B^{2r+1}))$$

$$= \sum_{r=1}^{20} \{\det(\text{adj} B^{2r+1}) \times \det(\text{adj} A^r)\}$$

$$= \sum_{r=1}^{20} \{\det(\text{adj} B)\}^{2r+1} \times \{\det(\text{adj} A)\}^r$$

$$= \sum_{r=1}^{20} (-1)^r = 0$$

6. Ans (A)

$$\text{Here } \vec{d} = \frac{\vec{a} + \vec{b} + \vec{c}}{\sqrt{3}}$$

$$|\vec{a} + \vec{b} + \vec{c} + \vec{d}|^2 = |\vec{a} + \vec{b} + \vec{c}|^2 + |\vec{d}|^2 + 2\vec{d} \cdot (\vec{a} + \vec{b} + \vec{c})$$

$$= 4 + 2\sqrt{3}$$

7. **Ans (B)**

Equation of plane containing L_1 is

$$P_1 + \lambda P_1 = 0 \quad \dots(1)$$

Equation of plane containing L_2 is

$$P_3 + \mu P_4 = 0 \quad \dots(2)$$

for some λ and μ equation (1) and (2)

are representing the same plane

$\Rightarrow$ equation of plane containing L_1 and L_2 is

$$-7x + 7y - 8z - 3 = 0$$

Let $Q(\lambda, \lambda, \lambda)$ be a point in the plane

$$\Rightarrow \lambda = -\frac{3}{8}$$

$$\text{so, required distance} = \sqrt{3} |\lambda| = \frac{3\sqrt{3}}{8}$$

MATHEMATICS

SEC-2

1. **Ans (A,B)**

Taking log both the sides

we get

$$\begin{aligned} \ell n \beta &= \lim_{n \rightarrow \infty} \left[\frac{\ell n \{ (1!) (2!) \dots (n!) \}}{n^2} - \alpha \ell n (n) \right] \\ &= \lim_{n \rightarrow \infty} \left[\frac{\ell n 1 + (\ell n 1 + \ell n 2) + (\ell n 1 + \ell n 2 + \ell n 3) + \dots}{n^2} - \alpha \ell n (n) \right] \\ &= \lim_{n \rightarrow \infty} \left[\frac{n \ell n 1 + (n-1) \ell n 2 + (n-2) \ell n 3 + \dots}{n^2} - \alpha \ell n (n) \right] \\ &= \lim_{n \rightarrow \infty} \left[\frac{1}{n} \sum_{r=1}^{\infty} \left(\frac{n+1-r}{n} \right) \ell n \left(\frac{r}{n} \right) - \alpha \ell n (n) + \frac{n(n+1)}{2n^2} \ell n (n) \right] \\ &= \lim_{n \rightarrow \infty} \left[\frac{1}{n} \sum_{r=1}^n \left(\frac{n+1-r}{n} \right) \ell n \left(\frac{r}{n} \right) + \left(\frac{1}{2} + \frac{1}{2n} - \alpha \right) \ell n n \right] \end{aligned}$$

As limit exists

$$\alpha = \frac{1}{2} \text{ and } \ell n \beta = \int_0^1 (1-x) \ell n x dx = -\frac{3}{4}$$

$$\Rightarrow \beta = e^{-\frac{3}{4}}$$

2. **Ans (A,B)**

Given D.Eⁿ is

$$\begin{aligned} \frac{dy}{2x dx} + \frac{y}{x^2} &= \frac{\sin^{-1} x}{2x^2} \\ \Rightarrow \frac{dy}{dx} + \left(\frac{2}{x} \right) y &= \frac{\sin^{-1} x}{x} \end{aligned}$$

$$\text{I.F.} = e^{\int \frac{2}{x} dx} = e^{2 \ell n x} = x^2$$

solution is given by

$$y \times x^2 = \int \frac{\sin^{-1} x}{x} \times x^2 dx$$

$$\Rightarrow y \times x^2 = \frac{1}{2} (2x^2 - 1) \sin^{-1} x + \frac{1}{4} x \sqrt{1-x^2} + C$$

$$\text{at } x=0, y=0 \Rightarrow C=0$$

3. **Ans (A,B,C,D)**

$$f(x) = \begin{vmatrix} 5 + \sin^2 x & \cos^2 x & 4 \sin 2x \\ \sin^2 x & 5 + \cos^2 x & 4 \sin 2x \\ \sin^2 x & \cos^2 x & 5 + 4 \sin 2x \end{vmatrix}$$

Apply $C_1 \rightarrow C_1 + C_2 + C_3$

$$= (6 + 4 \sin 2x) \begin{vmatrix} 1 & \cos^2 x & 4 \sin 2x \\ 1 & 5 + \cos^2 x & 4 \sin 2x \\ 1 & \cos^2 x & 5 + 4 \sin 2x \end{vmatrix}$$

Apply $R_3 \rightarrow R_3 + R_1$

$$= (6 + 4 \sin 2x) \begin{vmatrix} 1 & \cos^2 x & 4 \sin 2x \\ 1 & 5 + \cos^2 x & 4 \sin 2x \\ 0 & 0 & 5 \end{vmatrix}$$

$$= (6 + 4 \sin 2x) 5 \times 5 = 25 \times (6 + 4 \sin 2x)$$

$$[50, 250]$$

4. Ans (A,B)

$$A = \cos\theta \cdot \cos 2\theta \cdot \cos 3\theta \dots \cos(n-1)\theta$$

$$B = \sin\theta \cdot \sin 2\theta \cdot \sin 3\theta \dots \sin(n-1)\theta$$

$$\text{where } \theta = \frac{2\pi}{2019}$$

$$n = 2019$$

$$2^{1009} \times A \times B = (2\sin\theta \cdot \cos\theta) (2\sin 2\theta \cdot \cos 2\theta)$$

$$\dots (2\sin(n-1)\theta \cdot \cos(n-1)\theta)$$

$$= \sin 2\theta \cdot \sin 4\theta \cdot \sin 6\theta \dots \sin(2n-2)\theta$$

$$\Rightarrow 2^{1009} \times A \times B = B$$

$$A = \frac{1}{2^{1009}} = 2^{-1009}$$

$$\frac{p}{q} = \frac{1009}{1} \Rightarrow p + q = 1010$$

5. Ans (A,B,C)

$$\sin x + \sin y = 1$$

$$\Rightarrow \cos x + \cos y \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{\cos x}{\cos y}$$

$$\Rightarrow -\sin x - (\sin y) \left(\frac{dy}{dx} \right)^2 + (\cos y) \frac{d^2y}{dx^2} = 0$$

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{\sin x + \sin y \left(-\frac{\cos x}{\cos y} \right)^2}{\cos^2 y \cdot \sin x + \sin y \cdot \cos^2 x}$$

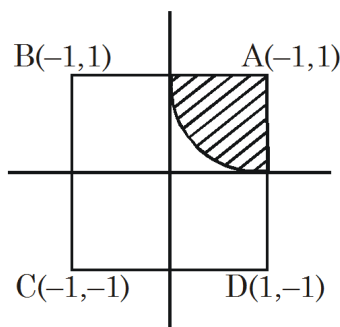
$$= \frac{(\cos y)^3}{(\cos y)^3}$$

$$\text{Put } \sin x = t, \sin y = 1-t, \cos y = \sqrt{2t-t^2}$$

$$\lim_{x \rightarrow 0} x^\alpha \frac{d^2y}{dx^2} = \lim_{t \rightarrow 0} \frac{t^\alpha [(2t-t^2)t + (1-t)(1-t^2)]}{(\sqrt{2t-t^2})^3}$$

$$\text{exists} \Rightarrow \alpha \geq \frac{3}{2}$$

6. Ans (B,C)



$$\text{Required area} = 4 \times \text{area in quadrant 1}$$

$$= 4 \times \left(1 - \frac{\pi}{4} \right) = 4 - \pi$$

7. Ans (A,B)

$$\lim_{x \rightarrow 0} \frac{x^2 \frac{\sin \beta x}{\beta x} \times \beta x}{\alpha x + \frac{(\alpha x)^3}{3} + \dots - \left(x - \frac{x^3}{3} + \dots \right)} = 4$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{x^3 \times \beta}{x(\alpha - 1) + \left(\frac{\alpha^3}{3} + \frac{1}{3!} \right) x^3 + \dots} = 4$$

$$\Rightarrow \alpha = 1 \text{ and } \frac{\beta}{\frac{1}{3} + \frac{1}{3!}} = 4$$

$$\Rightarrow \beta = 4 \times \left(\frac{1}{3} + \frac{1}{6} \right)$$

$$= 4 \times \frac{1}{2} = 2$$

MATHEMATICS

SEC-3

1. Ans (C)

using V.T.P

$$\text{we get } \vec{r} \cdot \vec{a} = \vec{r} \cdot \vec{b} = \vec{r} \cdot \vec{c}$$

$$\Rightarrow \vec{r} = \pm \frac{1}{\sqrt{3}} (\vec{a} + \vec{b} + \vec{c})$$

2. Ans (A)

$$\vec{r} = 12\vec{a} + 5\vec{b} + 13\vec{c}$$

$$\lambda = \vec{r} \cdot \left(\frac{\vec{a} + \vec{b} + \vec{c}}{\sqrt{3}} \right) = \frac{30}{\sqrt{3}}$$

3. Ans (B)

$$\frac{a+b+c}{b} = 2 \left(\frac{f(1)}{f(1)-f(-1)} \right)$$

$$= 2 \left(\frac{1}{1 - \frac{f(-1)}{f(1)}} \right)$$

$$\text{minimum of } \frac{a+b+c}{b} = 2 \text{ as } \frac{f(-1)}{f(1)} \geq 0$$

4. Ans (B)

$$H(x) = (x-\alpha)(x-1)(x-2)(x-3) + x$$

$$\Rightarrow H(-1) = (-1-\alpha)(-2)(-3)(-4) - 1 = 24(1+\alpha) - 1$$

$$H(5) = (5-\alpha)4 \times 3 \times 2 + 5 = 24(5-\alpha) + 5$$

$$H(-1) + H(5) = 148$$

$$\text{Similarly } H(0) + H(4) = 28$$